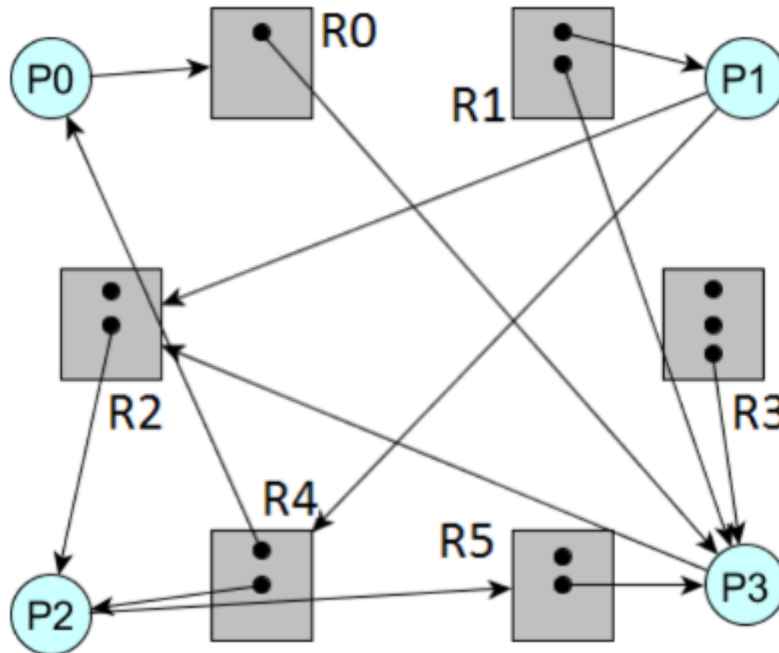


Assignment 2

Question 1:

A Resource Allocation Graph is given below:



Is there a deadlock in this Resource Allocation Graph? If there is a deadlock, briefly explain the reasons for it.

Question 2:

There are five processes (P0, P1, P2, P3, P4) and five shared resources (R0, R1, R2, R3, R4) arranged in a circular manner. Each process P_i needs to acquire two adjacent resources simultaneously to perform its work. After completing its work, the process releases both resources. The processes keep repeating this behavior. design a simple synchronization solution that prevents deadlock. Also, briefly explain (in 2–3 lines) how your solution avoids deadlock.

Question 3:

Consider a memory system with the following free partition sizes (in KB, in the order they appear in memory):

100 KB, 500 KB, 250 KB, 150 KB, 600 KB

Using the dynamic storage allocation techniques (First Fit, Best Fit, and Worst Fit), place the following sequence of processes in the order they arrive:

P1(200 KB), P2(350 KB), P3(100 KB), P4(450 KB), P5(150 KB)

For each allocation strategy determine:

- Which processes get allocated successfully and which ones cannot be allocated?
- The final state of the free memory partitions after all allocation attempts.
- Calculate the total external fragmentation (if any) and internal fragmentation (if any) remaining in the system.

Question 4:

A process has a virtual address space of 2^{16} bytes (64 KB) and the system uses demand paging with a page size of 2 KB (2048 bytes). The process is allocated only 6 physical frames in main memory.

The process generates the following page reference string:

1, 2, 3, 0, 1, 4, 0, 1, 2, 5, 1, 2, 3, 4, 5, 0, 1, 2

- a) How many virtual pages does the process have? How many bits are used for the page number and page offset?
- b) Simulate the page replacement using the FIFO algorithm with 6 frames. Count the total number of page faults.
- c) Simulate the same reference string using the LRU algorithm with 6 frames. Count the total number of page faults.
- d) Which algorithm performs better in this case? Briefly explain why.

Submission Guidelines

- Show all calculations clearly.
- Provide concise but precise explanations.
- Plagiarism will result in zero marks.
- This is an individual assignment and must be handwritten.
- Submissions will only be accepted in my presence. Any submission made in my absence will not be considered valid and will result in zero marks.
- **Deadline: 4 May 2026.**
- No submissions will be accepted after the deadline, as this is the last day of class and my office hours.